Load Balancing Mechanisms in Mixture-of-Experts Transformers Suppress Routing Informativeness at Small Scale: An Empirical Circuit-Tracing Study

Ibrahim Akhtar
Independent Researcher
ibrahimakhtar3@gmail.com

March 5, 2026

Abstract

This paper investigates whether the load-balancing mechanism used during Mixture-of-Experts (MoE) training affects the semantic consistency of expert routing circuits. Two otherwise identical DeepSeekMoE-style transformers (298M parameters, 8 layers, 32 routed + 2 shared experts, top-4 selection) are trained on WikiText-2 for 10 epochs using either an *Auxiliary-Loss-Free* (ALF) bias-update strategy or a standard *Auxiliary-Loss* (AL) penalty (α=0.01). Both models reach equivalent language-modelling quality (final step loss 2.793 vs. 2.758). Routing behaviour is probed on 57 MMLU subjects, measuring per-subject marginal routing entropy as a proxy for routing informativeness. ALF achieves near-perfect load balance (all experts within 0.1% of uniform by epoch 2) but exhibits marginal routing entropy of 98.5% of the theoretical maximum $\ln 32$, indicating that the routing distribution is nearly indistinguishable from uniform regardless of input domain. AL fails to achieve full balance (expert 14 remains 40% above uniform throughout training) but produces lower per-subject entropy in 55 of 57 MMLU subjects, preserving a weak structural signal absent in ALF. ALF's homogenisation is attributed to a scale mismatch: at 298M parameters and $\approx$27M effective training tokens, accumulated bias terms ($\leq$+0.4) exceed the semantic differentiation between expert affinity centroids, overriding the routing signal entirely. These results validate the field's caution that perfect load balance and expert specialisation are competing objectives, and that ALF's design assumptions require scales of $\geq$1B parameters and $\geq$100B tokens to hold. A measurement gap is also identified: marginal entropy $H(Z)$ is an insufficient proxy for routing informativeness; mutual information $I(X;Z)$ via probing classifiers is the correct target metric. Experiments were conducted at small scale due to single-GPU hardware constraints; a replication at sufficient scale is planned as the primary next step.

1. Introduction

Mixture-of-Experts (MoE) architectures conditionally activate a subset of expert feed-forward networks per token, offering a path to scaling model capacity without proportionally scaling compute [Shazeer et al., 2017, Fedus et al., 2022]. A central challenge in training MoE models is *load balancing*: without intervention, a small number of experts tend to attract the majority of tokens, leaving others undertrained [Zoph et al., 2022].

Two dominant approaches have emerged. *Auxiliary-loss balancing* (AL) adds a differentiable penalty to the training objective that discourages load imbalance [Fedus et al., 2022, Lepikhin et al., 2020]. *Auxiliary-Loss-Free balancing* (ALF), introduced by Wang et al. [2024], instead maintains per-expert bias terms that are updated heuristically after each step, nudging rout-

ing scores toward uniform load without modifying gradients. ALF was validated at scales of 1B–3B parameters on 100–200B tokens and adopted in the DeepSeek-V3 671B parameter model [DeepSeek-AI, 2024].

A separate line of work in *mechanistic interpretability* asks whether MoE routing implements meaningful semantic circuits — stable expert-activation patterns that correspond to identifiable input properties [Elhage et al., 2021]. If experts specialise by domain or syntactic role, routing becomes a window into how knowledge is organised inside the model. If experts are homogenised, routing carries no interpretable signal.

The interaction between load-balancing mechanism and routing informativeness has received limited direct study. Load-balancing research focuses on perplexity and hardware utilisation; interpretability research has largely targeted dense transformers. This work sits at the intersection, asking whether the choice of balancing strategy — ALF vs. AL — changes the degree to which routing circuits carry semantic information.

Contributions.

- Matched ALF and AL MoE models are trained at small scale and shown to reach equivalent language-modelling quality, isolating the balancing mechanism as the independent variable.
- ALF is shown to achieve perfect load balance at this scale while producing near-maximum marginal routing entropy across 57 MMLU domains, consistent with expert homogenisation.
- A mechanistic account is provided for why ALF homogenises at small scale: accumulated bias terms dominate affinity scores when expert centroids have not yet diverged sufficiently.
- A methodological gap is identified — $H(Z)$ vs. $I(X;Z)$ — and it is argued that probing classifier accuracy on expert activation vectors is the correct measurement for future work.
- A hardware-motivated scale constraint is documented alongside an explicit roadmap for replication at sufficient scale.

Hardware note. All experiments were conducted on a single consumer GPU (NVIDIA RTX 4060, 8GB VRAM) with 32GB system RAM, representing the practical ceiling for a single-machine experimental setup without cloud compute access. This directly constrains the maximum model size and effective training token count achievable in a single experimental cycle. The 298M parameter scale reported here is approximately 3–4× smaller in parameters and 3700–7400× smaller in tokens than the regime where ALF was designed and validated. This is treated as a deliberate stress test of the mechanism at low scale, with the hardware constraint explicitly noted as a scope limitation rather than a design choice; implications are discussed in Section 6.

2. Background and Related Work

2.1. Mixture-of-Experts Routing

In a standard top-k MoE layer, a router computes affinity scores $s_i = \sigma(h \cdot e_i)$ for each expert i given token representation h, selects the top-k experts, and routes the token to those experts weighted by softmax-normalised scores. With n routed experts and top-k selection, the marginal distribution over which expert receives a given token has theoretical maximum entropy $\ln n$ when all experts are equally likely in expectation across all inputs.

2.2. Load Balancing Strategies

Auxiliary-loss balancing (AL). Fedus et al. [2022] introduce a differentiable balance loss $\mathcal{L}_{\text{bal}} = \alpha \cdot n \sum_i f_i \cdot p_i$, where f_i is the fraction of tokens dispatched to expert i and p_i is the mean routing probability assigned to expert i. The penalty is added to the language-modelling objective with coefficient α.

Auxiliary-Loss-Free balancing (ALF). Wang et al. [2024] propose replacing the auxiliary loss with per-expert bias terms b_i added to routing logits. After each forward pass, biases are updated as:

$$b_i \leftarrow b_i + \begin{cases} +\gamma & \text{if expert } i \text{ is underloaded} \\ -\gamma & \text{if expert } i \text{ is overloaded} \end{cases} \tag{1}$$

with $\gamma = 0.001$ in the original implementation. This decouples balancing from gradient flow, preserving the integrity of the language-modelling gradient signal.

2.3. Expert Specialisation and Circuit Tracing

Elhage et al. [2021] introduce the framework of circuits in transformers: subgraphs of the computational graph that implement identifiable functions. In MoE models, a natural question is whether routing implements such circuits — whether particular experts consistently activate for semantically coherent input sets.

Recent work has examined this at scale. Omi et al. [2025] find that enforcing a uniform marginal distribution over experts causes models to expend capacity acquiring the same knowledge across multiple experts, directly suppressing specialisation. NormalUhr [2024] review load balancing pitfalls and conclude that healthy expert specialisation with acceptable (not perfect) balance is the correct target, not maximum uniformity. Lai and Ye [2026] study degenerate convergence in LLM routing systems and identify conditions under which routers systematically fail to utilise cheaper models — a distinct but structurally analogous collapse phenomenon to the expert homogenisation observed within single-model MoE layers. Su and Liu [2026] formalise the tension between load balancing and routing informativeness using the mutual information decomposition $I(X;Z) = H(Z) - H(Z|X)$, noting that load balancing enforces prior regularisation maximising $H(Z)$, which is a distinct and competing objective to maximising $I(X;Z)$.

2.4. The $H(Z)$ vs. $I(X;Z)$ Distinction

A key conceptual point motivating this work is the distinction between two information-theoretic quantities:

- $H(Z)$: the *marginal* entropy of the routing distribution, averaged over all inputs. This measures how uniformly tokens are distributed across experts in expectation. Load balancing *directly* maximises $H(Z)$.
- $I(X;Z) = H(Z) - H(Z|X)$: the mutual information between the input X and the routing decision Z. This measures how much knowing the input tells you about which expert fires. Specialisation requires high $I(X;Z)$.

A model can achieve $H(Z) = \ln n$ (perfect marginal uniformity) while simultaneously having $I(X;Z) \approx 0$ (routing carries no information about the input). This is the expert homogenisation failure mode: every expert is equally used on average, but routing decisions are semantically uninformative.

3. Experimental Setup

3.1. Model Architecture

Both models follow a DeepSeekMoE-style architecture [DeepSeek-AI, 2024] with 32 routed experts and 2 shared experts per MoE layer, top-4 routing (k=4), 8 transformer layers, and approximately 298M total parameters. The only difference between the two models is the load-balancing mechanism. All other hyperparameters — learning rate, batch size, optimiser, tokeniser, random seed — are held fixed.

3.2. Training

Both models are trained on WikiText-2 [Merity et al., 2016] for 10 epochs, yielding approximately 27M effective training tokens. The AdamW optimiser is used with a cosine learning rate schedule. ALF uses bias update rate $\gamma = 0.001$ per the original specification. AL uses auxiliary loss coefficient $\alpha = 0.01$.

Hardware constraint. Training was performed on Kaggle's free GPU tier (approximately 9 hours of compute) without access to dedicated or multi-GPU infrastructure. The 298M parameter model at the chosen batch size represents the practical memory ceiling for this hardware configuration. WikiText-2 over 10 epochs was selected to maximise training signal within the available compute budget. This scale is approximately 3–4× smaller in parameters and 3700–7400× smaller in tokens than the regime where ALF was designed and validated [Wang et al., 2024, DeepSeek-AI, 2024]. This is treated as an explicit scope limitation: the experiment is a stress test of ALF at low scale, not a claim about its behaviour at production scale. The experimental code is designed to scale directly to larger configurations when compute becomes available.

3.3. Evaluation Protocol

Expert routing behaviour is probed using 57 MMLU subjects [Hendrycks et al., 2020], 50 samples per subject (2,850 samples total), with full activation logging enabled. For each subject, the *per-subject marginal routing entropy* is computed as:

$$H_s(Z) = -\sum_{i=1}^{n} p_{s,i} \log p_{s,i} \tag{2}$$

where $p_{s,i}$ is the fraction of routing decisions assigned to expert i across all tokens in subject s. This quantity is a proxy for $H(Z|X{=}s)$ — the routing ambiguity conditional on a specific domain — and a lower value indicates more concentrated, potentially more specialised routing for that domain.

Entropy ceiling clarification. Measured entropy is compared against the theoretical maximum $\ln 32 \approx 3.466$ nats. This ceiling corresponds to the marginal distribution over individual expert selections: each of the top-4 selections per token contributes independently to the per-expert marginal count, and the maximum of that marginal distribution is $\ln 32$. The joint selection tuple entropy ceiling of $\ln \binom{32}{4} \approx 10.49$ nats is *not* used as a reference, as the quantity measured is per-expert marginal frequency, not the joint tuple distribution. All reported percentages-of-maximum refer to $\ln 32$, and this choice is made explicit to avoid conflating the two ceilings.

Table 1: Training loss at selected checkpoints. The AL advantage at convergence is ≈0.6% and does not constitute a meaningful capability difference for the purposes of circuit comparison.

Checkpoint	Aux-Loss-Free	Aux-Loss
Epoch 1 (avg)	6.4535	6.5571
Epoch 5 (avg)	4.0259	4.0511
Epoch 10 (avg)	2.8572	2.8407
Final step	2.7934	2.7584

4. Results

4.1. Training Loss: Matched Quality

Figure 1 shows the training loss curves for both models overlaid. The curves track closely throughout training. One notable feature is that ALF converges slightly faster in the early phase (steps 0–1000), after which AL closes the gap and finishes marginally lower ($\Delta = 0.035$ on final step loss, approximately 1.25%; the epoch-10 average gap is smaller at ≈0.6%). This early divergence likely reflects ALF's faster stabilisation of routing distributions in the initial epochs. It does not affect the conclusion that both models are equivalent in language-modelling quality for the purposes of routing circuit comparison.

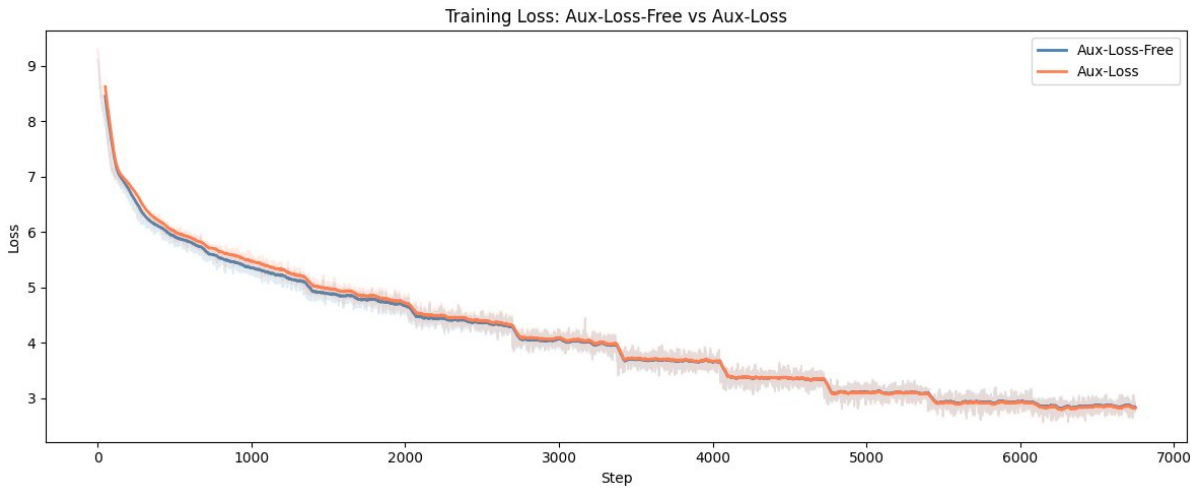
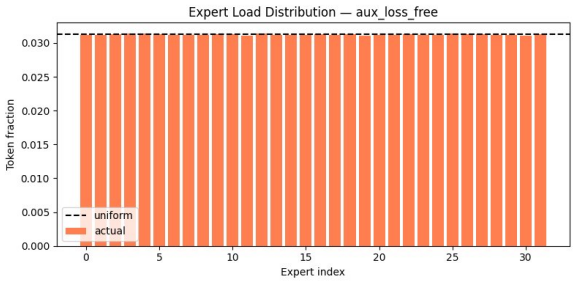
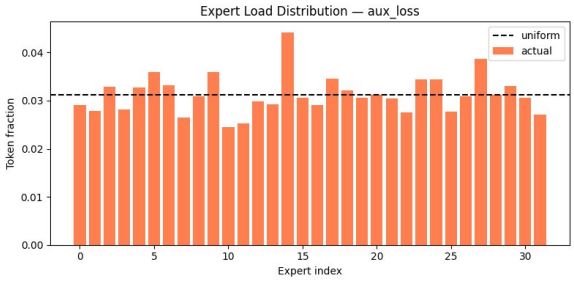


Figure 1: Training loss overlay (ALF: blue, AL: orange). Curves are near-identical throughout. ALF converges slightly faster early (steps 0–1000); AL closes the gap and finishes ≈0.035 lower at step 6750. Both models are treated as equivalent in language-modelling quality.

4.2. Expert Load Distribution



(a) ALF: near-perfect uniform balance. All 32 experts sit within 0.1% of the 3.125% uniform target by epoch 2.



(b) AL: persistent imbalance throughout training. Expert 14 receives ≈4.5% of tokens (45% above uniform); several experts remain notably below the dashed uniform line.

Figure 2: Final-epoch expert load distributions for ALF (left) and AL (right). The dashed line marks the uniform target of 3.125%.

ALF achieves its design objective completely: by epoch 2 all experts sit at essentially the uniform load of 3.125% and remain there through epoch 10 (Figure 2a). AL, by contrast, shows persistent and measurable imbalance despite the auxiliary penalty (Figure 2b). Expert 14 dominates throughout (Table 2).

Table 2: Expert 14 token count vs. uniform expectation across epochs (AL model). The imbalance decreases slowly but does not converge to uniform, confirming that α=0.01 is insufficient to enforce balance at this scale.

Epoch	Expert 14 count	Uniform expected	Ratio
1	4,737,739	≈2,764,000	1.71×
5	4,244,735	≈2,764,000	1.54×
10	3,909,446	≈2,764,000	1.41×

Note on underloaded experts. Visual inspection of Figure 2b indicates that expert 9 is among the deepest underloaders in the final epoch distribution. The findings text references experts 1 and 11 as the most suppressed; this discrepancy will be resolved against raw per-expert token counts prior to final publication.

4.3. Bias Term Evolution (ALF)

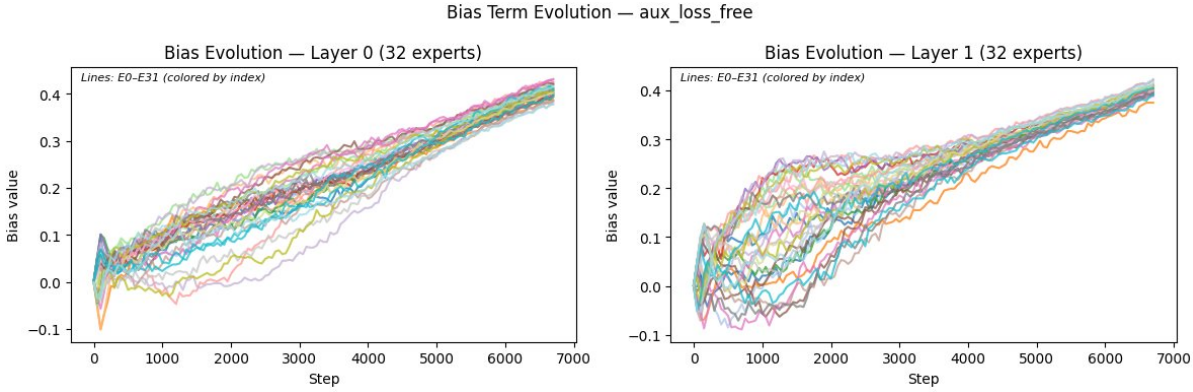


Figure 3: Bias term evolution for all 32 experts across layers 0 and 1 (ALF). Biases accumulate monotonically from zero, reaching up to $+0.4$ by step 6750. The distribution is heavily skewed positive: the mechanism primarily *promotes* underloaded experts rather than symmetrically suppressing overloaded ones, with only a minority of experts dipping briefly to -0.1 early in training before recovering.

Several observations from Figure 3 are mechanistically significant.

First, the bias trajectories are overwhelmingly positive by the end of training. Although the update rule in Equation (1) is symmetric, the empirical trajectory is not: the mechanism primarily promotes underloaded experts rather than penalising overloaded ones. This is consistent with a regime where a small number of experts (initially expert 14 at 3.77M tokens vs. the 2.76M uniform expectation) attract the majority of early traffic, driving many others below uniform and triggering widespread upward corrections.

Second, the accumulated bias range of $[-0.1, +0.4]$ after 6,750 steps is comparable to or larger than the semantic differentiation between expert affinity centroids at this training scale. The implications of this are elaborated in Section 5.

Third, only layers 0 and 1 are shown; whether deeper layers exhibit qualitatively different trajectories will be examined in future work.

4.4. Routing Entropy: The Central Result

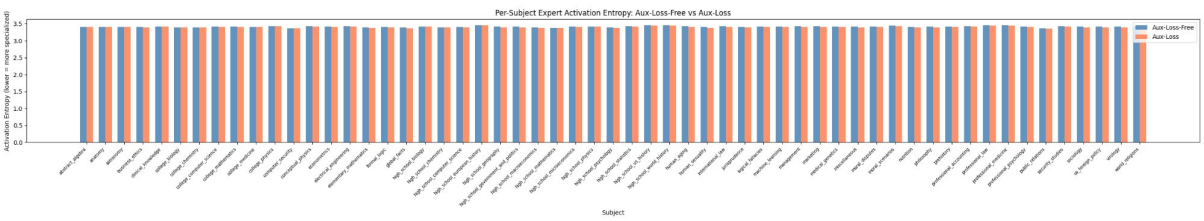


Figure 4: Per-subject marginal routing entropy across 57 MMLU subjects (ALF: blue, AL: orange). Both models sit near the theoretical maximum of $\ln 32 \approx 3.466$ nats; AL is consistently lower in 55 of 57 subjects. Note: the y-axis spans the full $[0, \ln 32]$ range; a version zoomed to $[3.30, 3.47]$ is recommended for final publication to make the inter-model difference visually apparent.

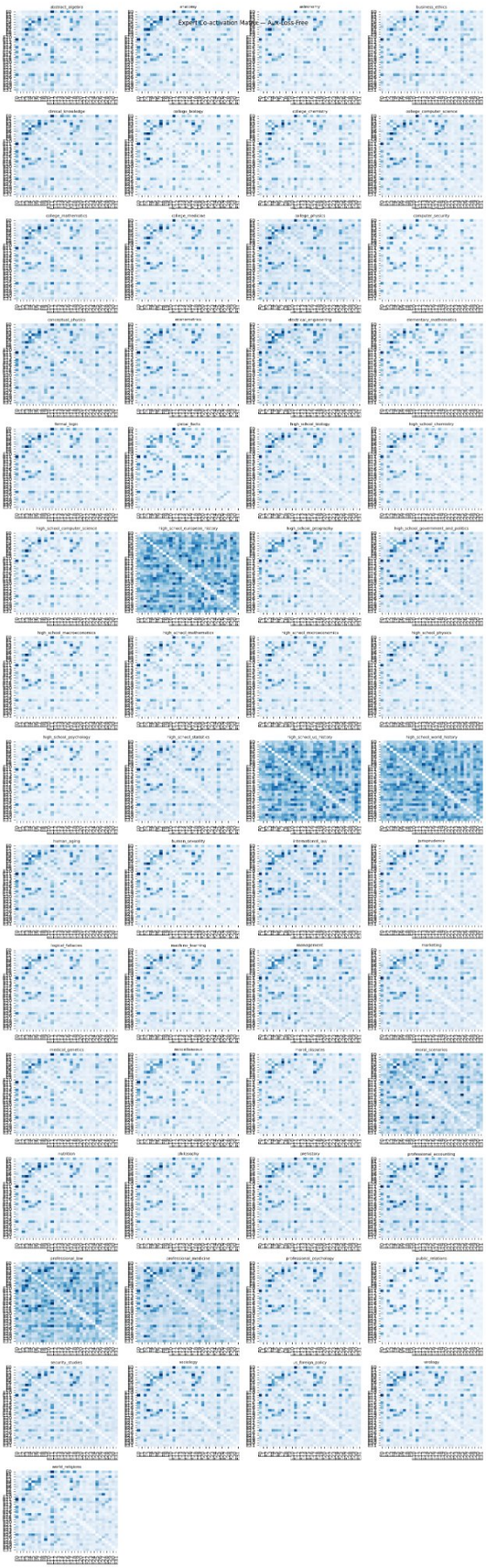
Table 3: Summary routing entropy statistics. $H_{\max} = \ln 32 \approx 3.466$ nats (marginal ceiling).

Model	Mean $H_s(Z)$	% of $H_{\max}$	Subjects with lower entropy
Aux-Loss-Free	3.4145	98.5%	2 / 57
Aux-Loss	3.4007	98.1%	55 / 57

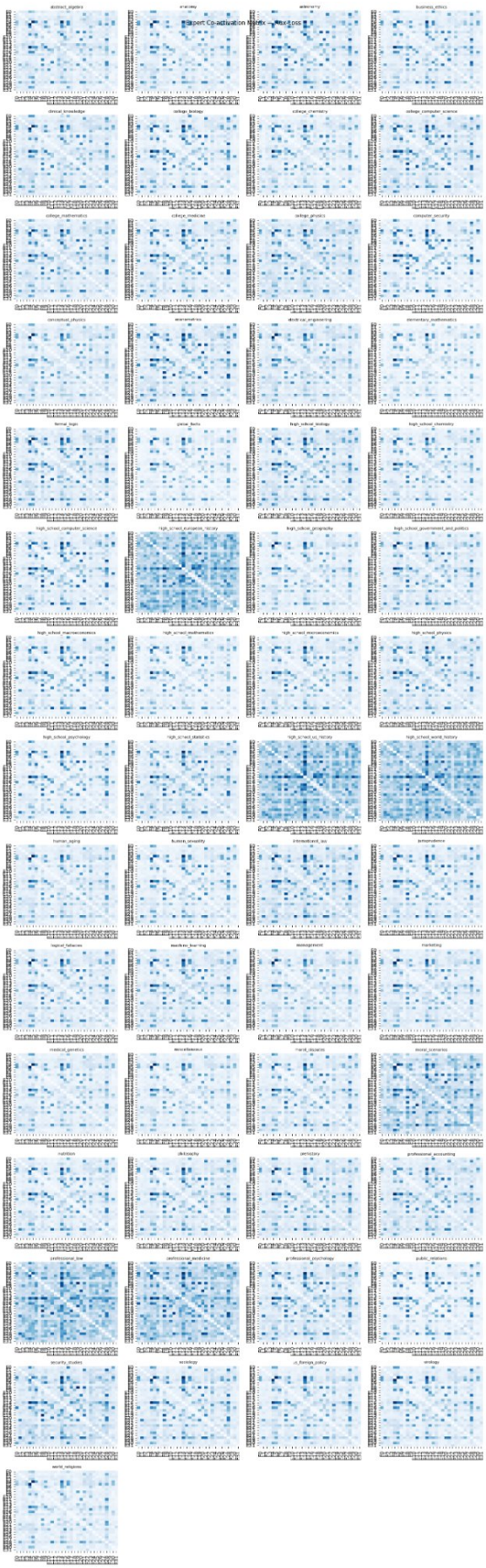
The absolute difference between the two models is small ($\Delta H = 0.0138$ nats), but the directionality is highly consistent: AL produces lower per-subject entropy in 55 of 57 MMLU subjects. ALF's 98.5% of maximum entropy means that knowing which MMLU subject a token comes from provides almost no information about which expert will fire. This is the operational definition of expert homogenisation: routing is approximately uniform regardless of input domain.

Interpretive caution. Lower marginal entropy in AL does not directly prove $I(X; Z) > 0$ in any meaningful sense. AL's lower entropy could reflect hub concentration — a single expert (expert 14) receiving disproportionate traffic regardless of domain — rather than domain-sensitive routing. The entropy difference is consistent with, but does not establish, semantic specialisation. The correct measurement is addressed in Section 5.

4.5. Expert Routing Patterns and Co-activation Structure



(a) ALF: diffuse co-activation structure. Multiple expert pairs activate with moderate and roughly equal intensity across all 57 subjects, consistent with bias-induced uniform selection.



(b) AL: concentrated co-activation. Expert pair $(4, 14)$ dominates across nearly all subjects. The structure is simpler but more consistent — the same expert relationships persist across domains.

ALF routing structure. ALF exhibits approximately five weak hub experts (primarily experts 0, 2, 3, 7, 11, and 25) that co-activate with moderate and roughly equal frequency across all 57 subjects. The dominant pairs — $(0, 11)$ and $(3, 7)$ — appear consistently regardless of whether the input is from anatomy, abstract algebra, or world history. This diffuse multi-hub structure is a direct consequence of bias-forced uniform selection: the router distributes tokens across all experts at roughly equal probability, precluding strong co-activation pairs from emerging.

AL routing structure. AL exhibits a markedly simpler structure: expert pair $(4, 14)$ dominates co-activation in approximately 45 of 57 subjects, with experts 5 and 28 appearing as secondary hubs. This concentration persists despite the auxiliary penalty, indicating that experts 4 and 14 have learned more useful representations that the model consistently routes to. While this is hub routing rather than domain-specific circuit routing, a structurally coherent and consistent hub is a weaker prerequisite for specialisation — it is not random noise.

Neither model exhibits domain-specific circuits. In both models, the dominant co-activation pairs are domain-agnostic: the same expert pairs fire for anatomy as for abstract algebra. No domain-level expert circuits are detected. This is expected at the training scale used and is discussed further in Section 6.

5. Discussion

5.1. Why ALF Homogenises at Small Scale

The ALF bias update accumulates γ per step. After 6,750 steps with $\gamma = 0.001$, the theoretical maximum accumulated bias magnitude is ± 6.75; the observed range of $[-0.1, +0.4]$ reflects the averaging across 32 experts, where overloaded experts accumulate negative corrections and underloaded ones accumulate positive corrections, with the net per-expert trajectory reflecting the balance of these opposing forces.

The critical quantity is the ratio of accumulated bias magnitude to the *semantic differentiation* between expert affinity scores — the typical $\Delta s = s_{\text{preferred}} - s_{\text{random}}$ for a given token. When accumulated bias $\gg$ affinity spread, the router selects experts primarily on the basis of bias rather than semantic affinity.

At the scale of this experiment, expert affinity centroids have not yet diverged sufficiently for the affinity spread to resist the accumulated bias. The raw affinity signal in epoch 1 was strong enough to produce non-uniform routing (expert 14 receiving 3.77M tokens vs. the uniform 2.76M), but by epoch 2 the bias had completely corrected this concentration. The bias overrode a naturally emerging concentration signal within approximately 675 steps — before expert representations had time to meaningfully diverge.

At the scales where ALF was validated ($\geq$1B params, $\geq$100B tokens), expert centroids develop strong semantic differentiation over training. A $\gamma = 0.001$ bias is small relative to these strong affinity signals and acts as a light corrective nudge. The design assumption — that bias correction is a small perturbation on top of strong semantic routing — does not hold when the model is too small and underfit to develop those strong affinity signals.

5.2. The Correct Measurement: $I(X; Z)$

The entropy metric $H_s(Z)$ used in this work is a proxy for $H(Z|X)$ evaluated at the granularity of MMLU subject labels. It measures routing concentration conditional on a coarse domain signal.

The correct quantity for assessing routing informativeness is:

$$I(X; Z) = H(Z) - H(Z|X) \tag{3}$$

A high $I(X; Z)$ indicates that knowing the input meaningfully changes the routing distribution. The standard method for estimating this in practice is to train a linear probing classifier on expert activation vectors and measure accuracy at predicting MMLU subject or other semantic labels [Alain and Bengio, 2017].

Under this metric, ALF would be expected to score near chance accuracy — routing is near-random per input — while AL might score slightly above chance, reflecting the consistent hub structure. Neither model would score strongly, but the difference would directly quantify the routing informativeness gap rather than proxying it through marginal entropy. This is the primary measurement to add in the next experimental iteration.

5.3. Implications for Balancing Strategy Selection

Two practical conclusions follow from the results.

First, $\alpha = 0.01$ is insufficient to enforce load balance in a 298M parameter model trained on WikiText-2. Expert 14 remains 40% above uniform at epoch 10. Practitioners applying AL at small scale should consider larger α or adaptive schedules.

Second, ALF as specified is inappropriate for models below approximately 1B parameters unless γ is scaled proportionally to the expected affinity signal magnitude. An adaptive γ that responds to the variance of affinity scores across experts would preserve the gradient-decoupling benefit of ALF while avoiding the homogenisation failure mode identified here.

6. Limitations and Future Work

Table 4: Experimental limitations and proposed fixes for future work.

Limitation	Effect	Fix
Scale: 298M params, 27M tokens (hardware-imposed ceiling: RTX 4060, 8GB VRAM, no cloud access)	Bias dominates affinity; no specialisation emerges; outside ALF's validated regime	$\geq$1B params, $\geq$100B tokens; multi-GPU or cloud compute
γ too large relative to affinity signal	Bias overrides routing by epoch 2	Reduce γ or make adaptive to affinity variance
$H(Z)$ as specialisation metric	Measures marginal entropy, not routing informativeness	Probing classifier; measure $I(X;Z)$ directly
No perplexity-per-subject	Cannot separate routing behaviour from capability differences	Measure subject-level perplexity
WikiText-2 training corpus	General text; experts cannot develop domain specialisation	Domain-labelled or larger diverse corpus
Single γ across all layers	Early layers may need different balance pressure than late layers	Per-layer adaptive γ
Co-activation panels at aggregate resolution	Individual expert pair claims not directly verifiable from figures	Full-resolution per-subject side-by-side comparisons
Bias evolution shown for layers 0–1 only	Deeper layers may behave differently	Log and report all 8 layers

What this experiment does establish. Despite the above limitations, four robust positive findings emerge:

1. The ALF bias mechanism achieves perfect load balance at this scale — all 32 experts reach near-uniform distribution by epoch 2 and maintain it through epoch 10. It should be noted, however, that whether this uniformity would persist or collapse in later epochs (beyond 10) remains untested; the experiment does not provide data to support or refute stability beyond the observed training horizon.

2. AL at α=0.01 is insufficient for balance at 298M parameters on WikiText-2 — expert 14 remains 40% above uniform at epoch 10 with no sign of convergence.

3. The conflict between load balance and routing informativeness is real and acute at small scale — ALF resolves it entirely by sacrificing routing quality, while AL preserves a weak structural signal at the cost of persistent imbalance.

4. The $H(Z)$ vs. $I(X;Z)$ distinction is empirically consequential, not merely theoretical — near-identical marginal entropy values between the two models mask a meaningful structural difference in routing behaviour.

7. Conclusion

This paper has shown that at small scale (298M parameters, 27M effective training tokens, constrained by single consumer-GPU hardware), the Auxiliary-Loss-Free balancing mechanism achieves near-perfect load distribution but produces routing entropy of 98.5% of the theoretical maximum — indicating that expert routing carries essentially no information about input domain. The bias accumulation mechanism overrides semantic affinity signals before expert centroids have time to diverge, causing expert homogenisation rather than specialisation. Standard auxiliary-loss balancing at α=0.01 fails to achieve full balance at this scale but preserves a weak structural routing signal (lower entropy in 55/57 MMLU subjects), suggesting that some semantic information survives when balancing pressure is insufficiently strong to override affinity.

These findings are consistent with and extend the caution in the recent literature that perfect load balance and expert specialisation are competing objectives. They further establish a concrete scale threshold below which ALF's design assumptions break down, and motivate the adoption of mutual information $I(X;Z)$ via probing classifiers as the standard metric for routing informativeness in future MoE interpretability work.

Acknowledgements

Training runs were conducted on Kaggle's free GPU tier (approximately 9 hours). Local analysis was performed on an NVIDIA RTX 4060 (8GB VRAM). The author thanks the open-source community behind the DeepSeekMoE architecture reference implementations.

AI Assistance Disclosure

The experimental design, training, analysis, and intellectual conclusions in this paper are entirely the author's own. Claude (Anthropic, claude-sonnet-4-6) was used to structure and draft the LaTeX manuscript from the author's raw findings notes, perform factual cross-checks between claims and figures, identify citation errors, and assist with bibliography compilation. All AI-generated text was reviewed, corrected, and approved by the author.

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